

SAE Alarme

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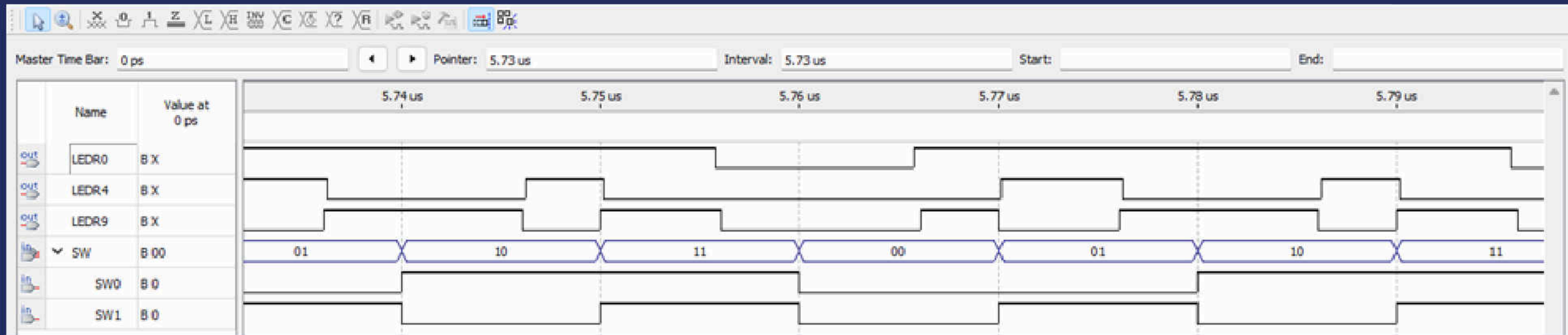
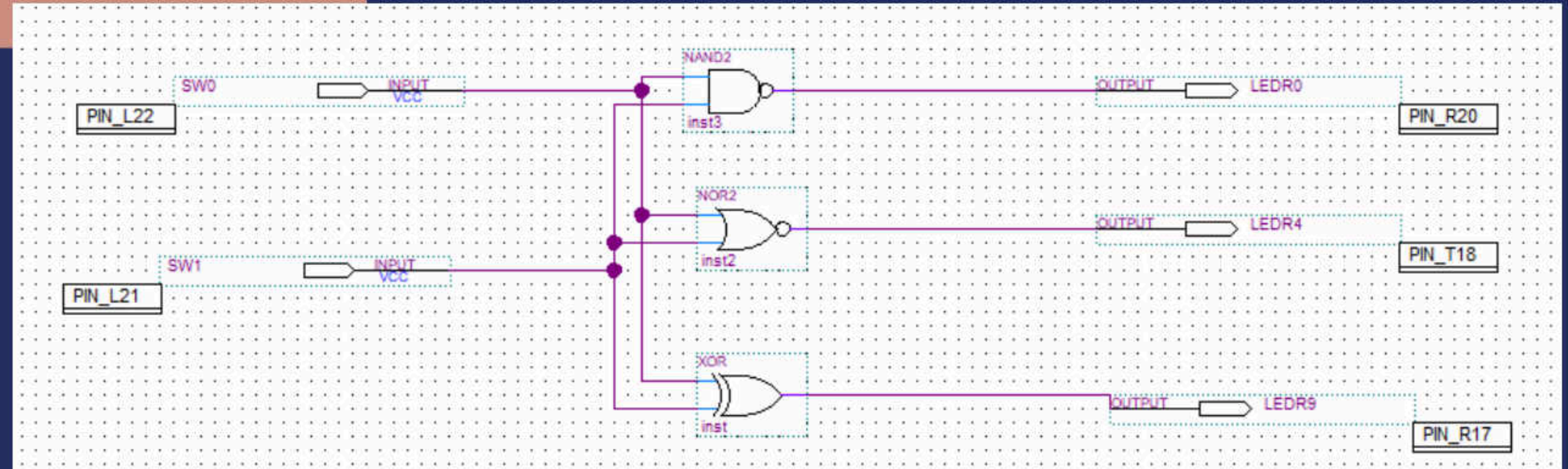
1. Présentation of SAE

Design → secure access control system

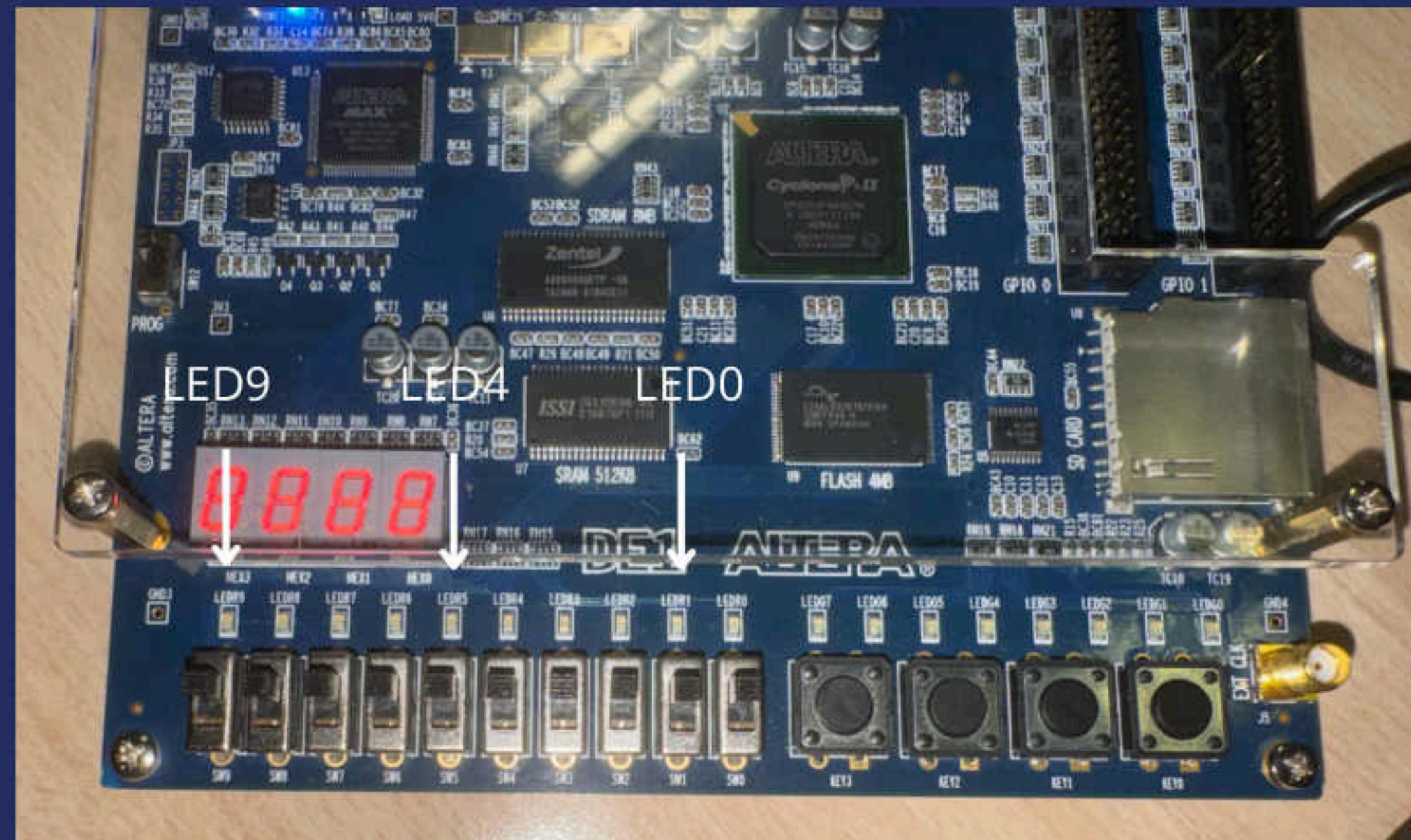
Create → logic circuit with Quartus II capable of:

1. Save an access code
2. Verify a user code
3. Open the door if the code is correct
4. Activate an alarm if the code is false

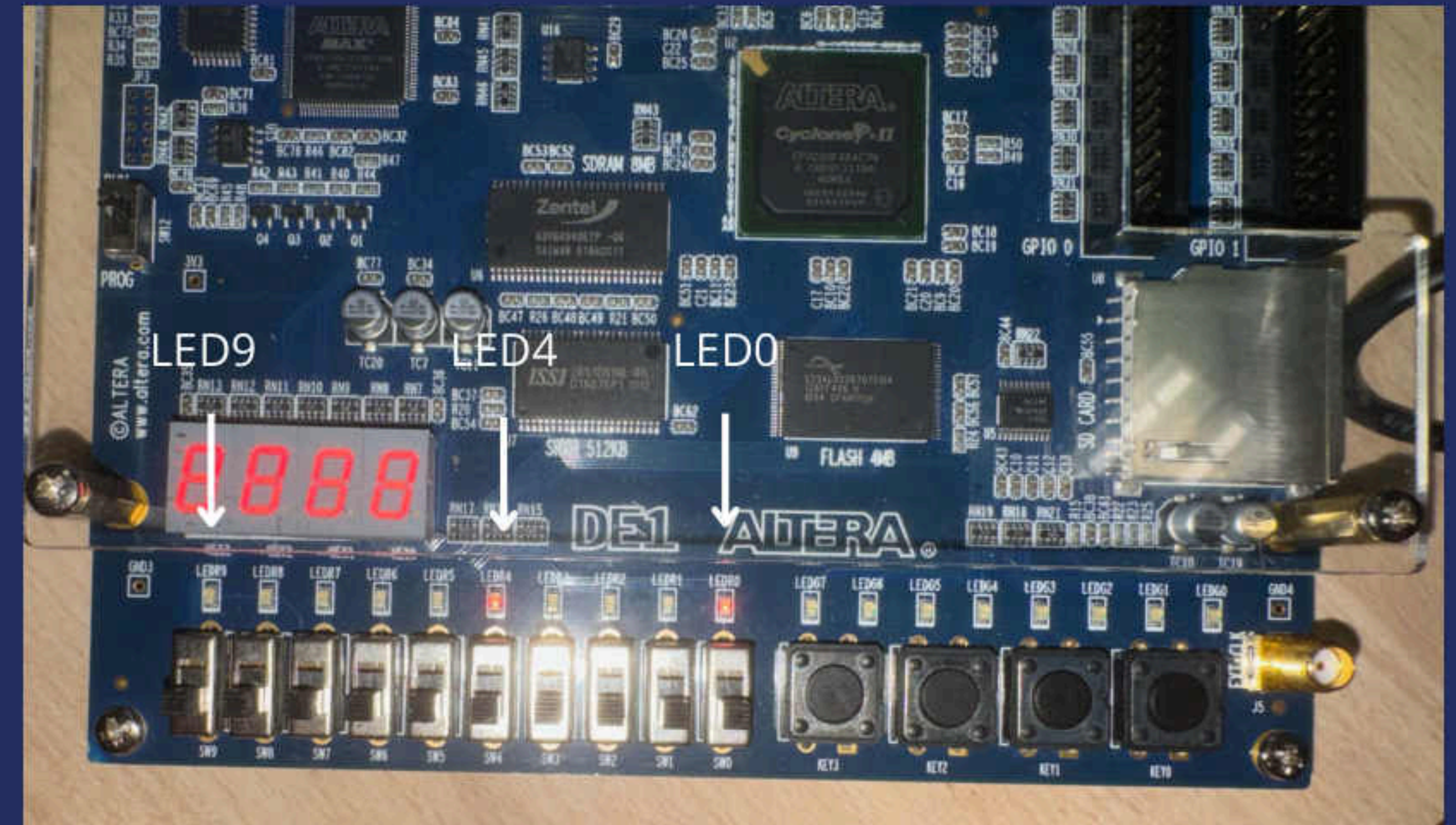
2. Test Quartus II



2. Test Quartus II



SW1 SW0



SW1 SW0

3. Bloc Alarme

GO.C	GO.M	I ₀	I ₁	SA	SP
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	0	1	1	0
1	0	1	0	0	0
1	0	1	1	1	0
1	1	0	0	1	0
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	1	1	0

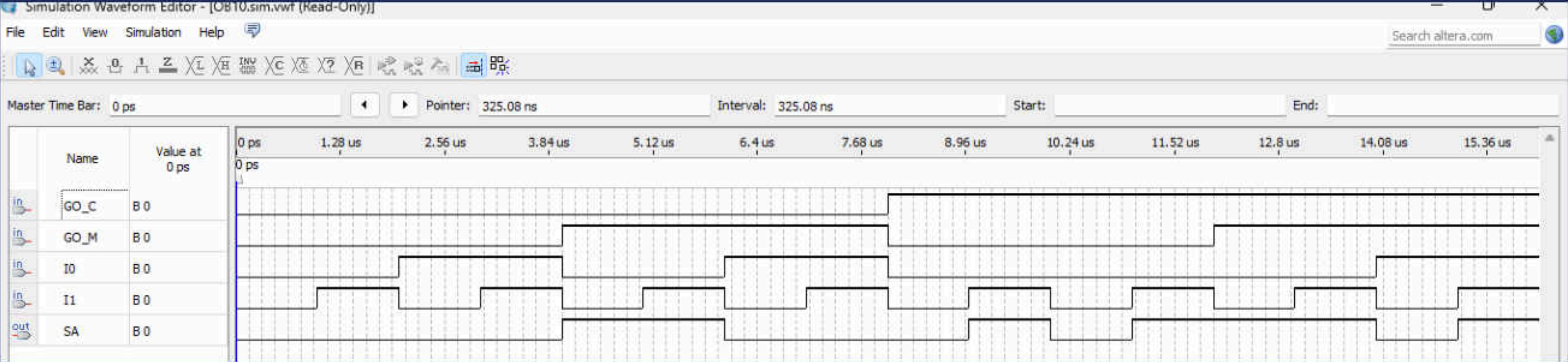
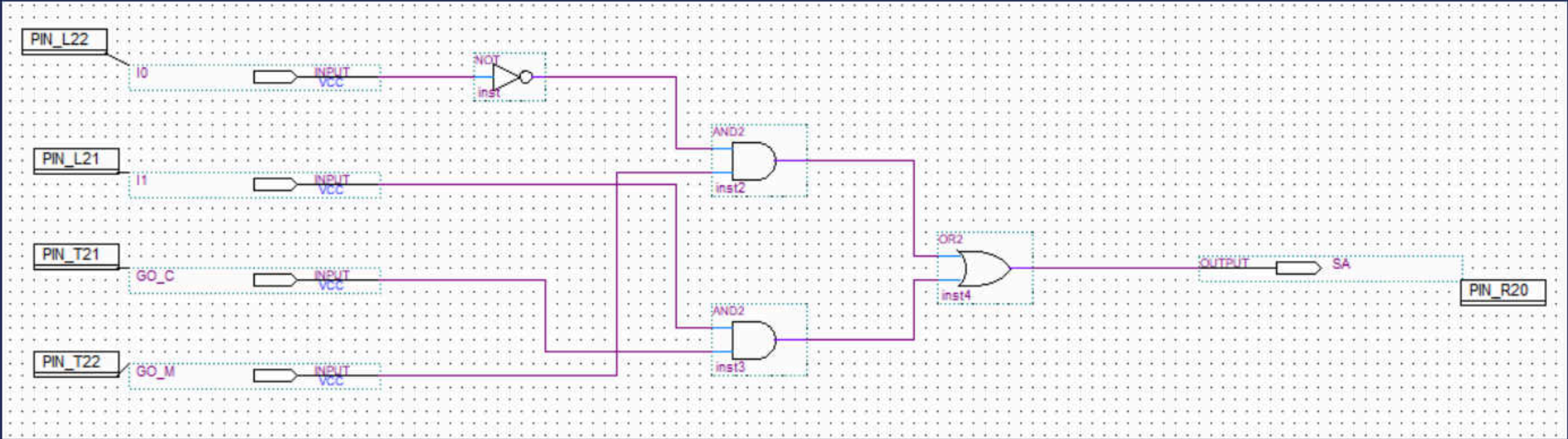
Table de vérité

		00	01	11	10
SA	GO.C.GOH	00	01	11	10
	I ₀ .I ₁	00	01	11	10
	00	0	1	1	0
	01	0	1	1	1
	11	0	0	1	1
	10	0	0	0	0

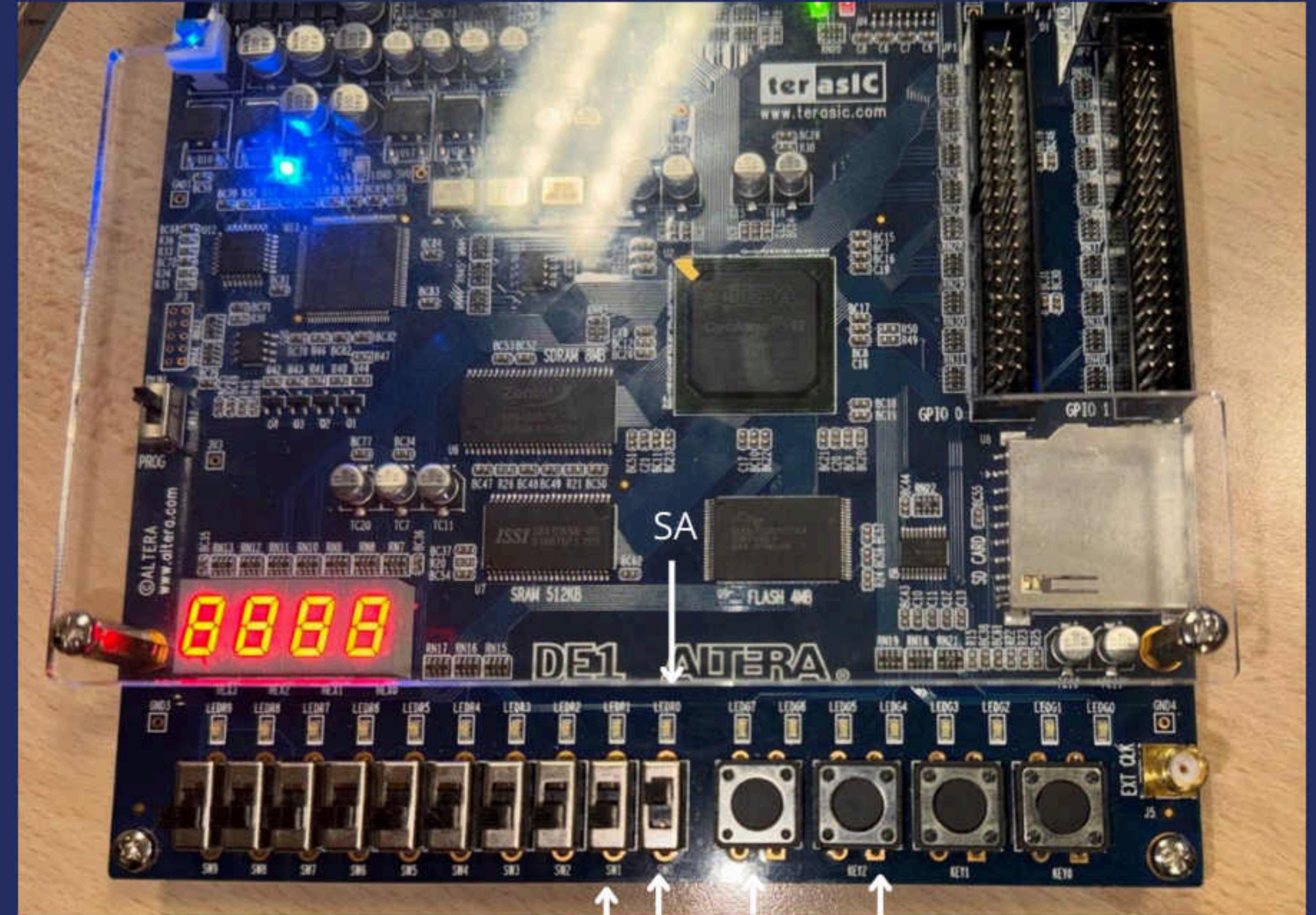
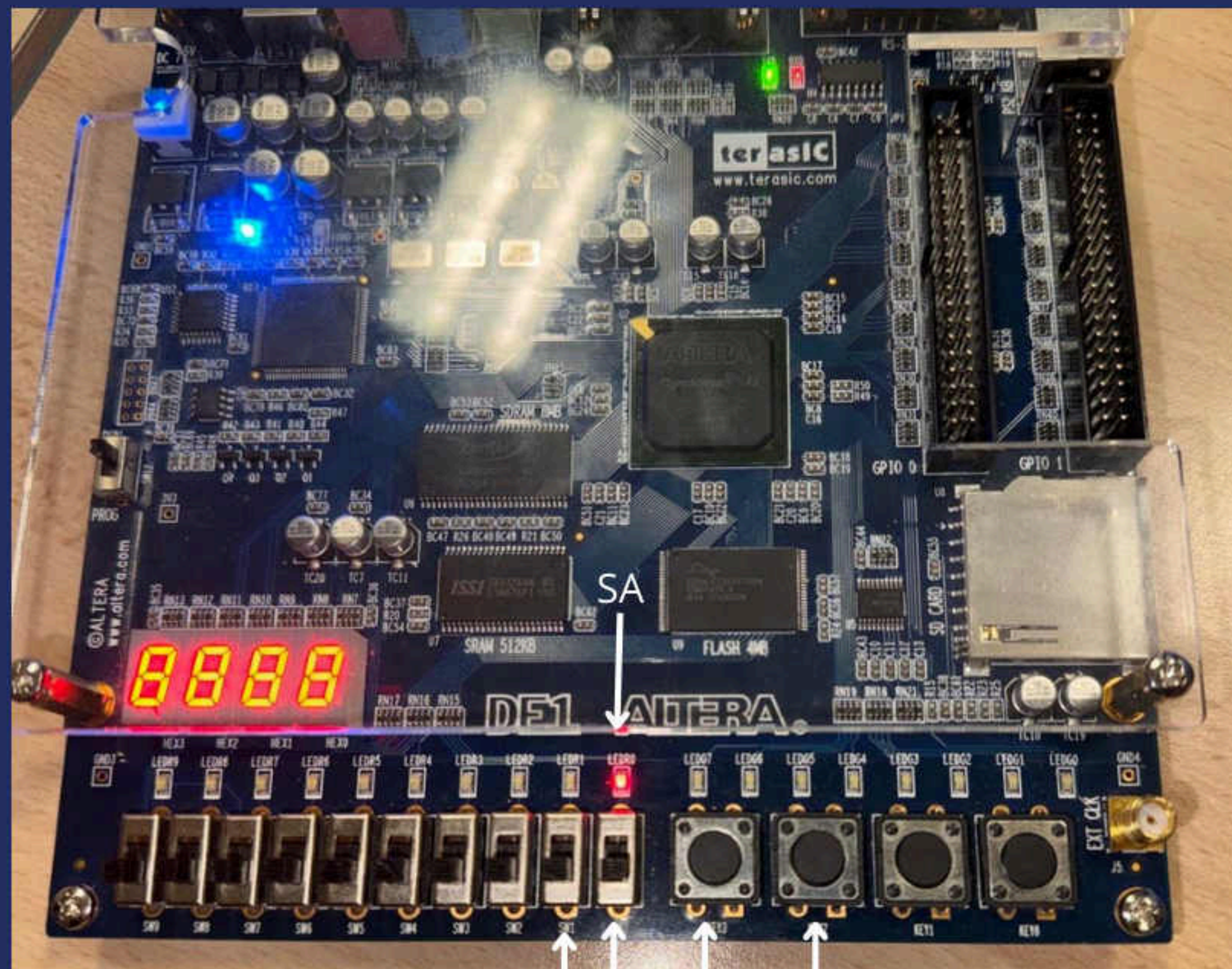
$SA = GOH \cdot \overline{I_0} + GOH \cdot I_1$

Table de karnaugh

3. Bloc Alarme



3. Bloc Alarme



3. Bloc Porte

G.O.C	G.O.M	I_0	I_1	SA	SP
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	0	1	1	0
1	0	1	0	0	0
1	0	1	1	1	0
1	1	0	0	1	0
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	1	1	0

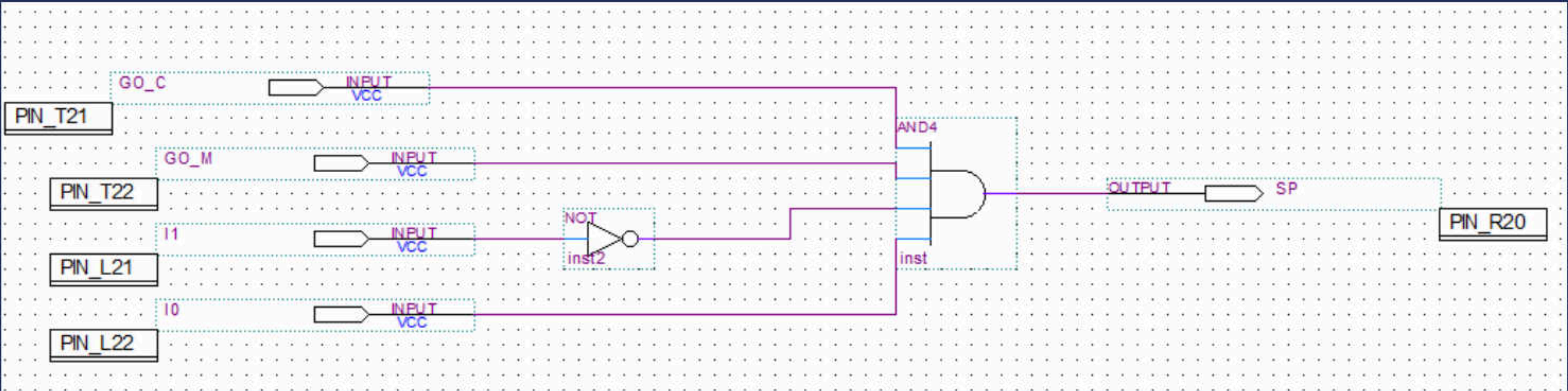
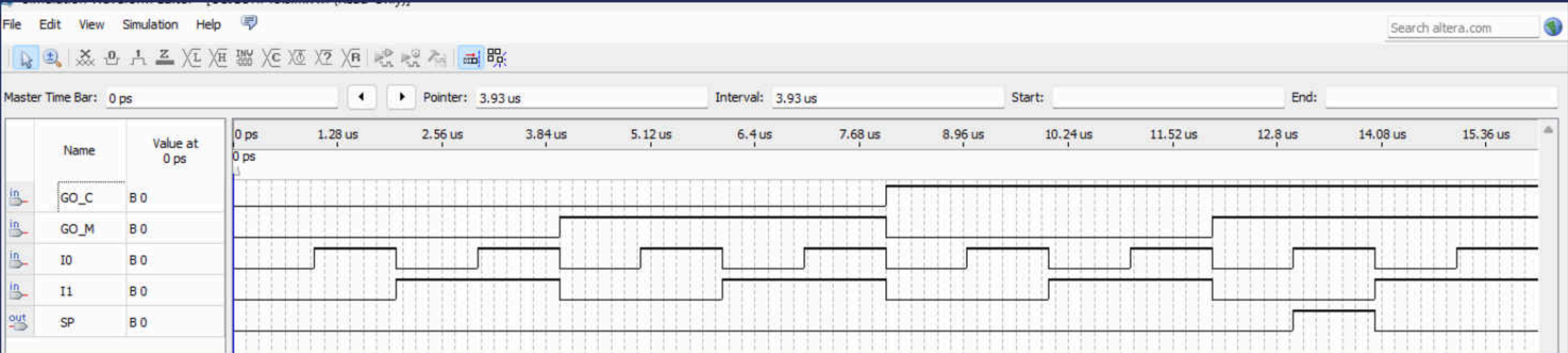
Table de vérité

SP		G.O.C.G.O.M			
		$I_0.I_1$			
		00	01	11	10
00		0	0	0	0
01		0	0	0	0
11		0	0	0	0
10		0	0	1	0

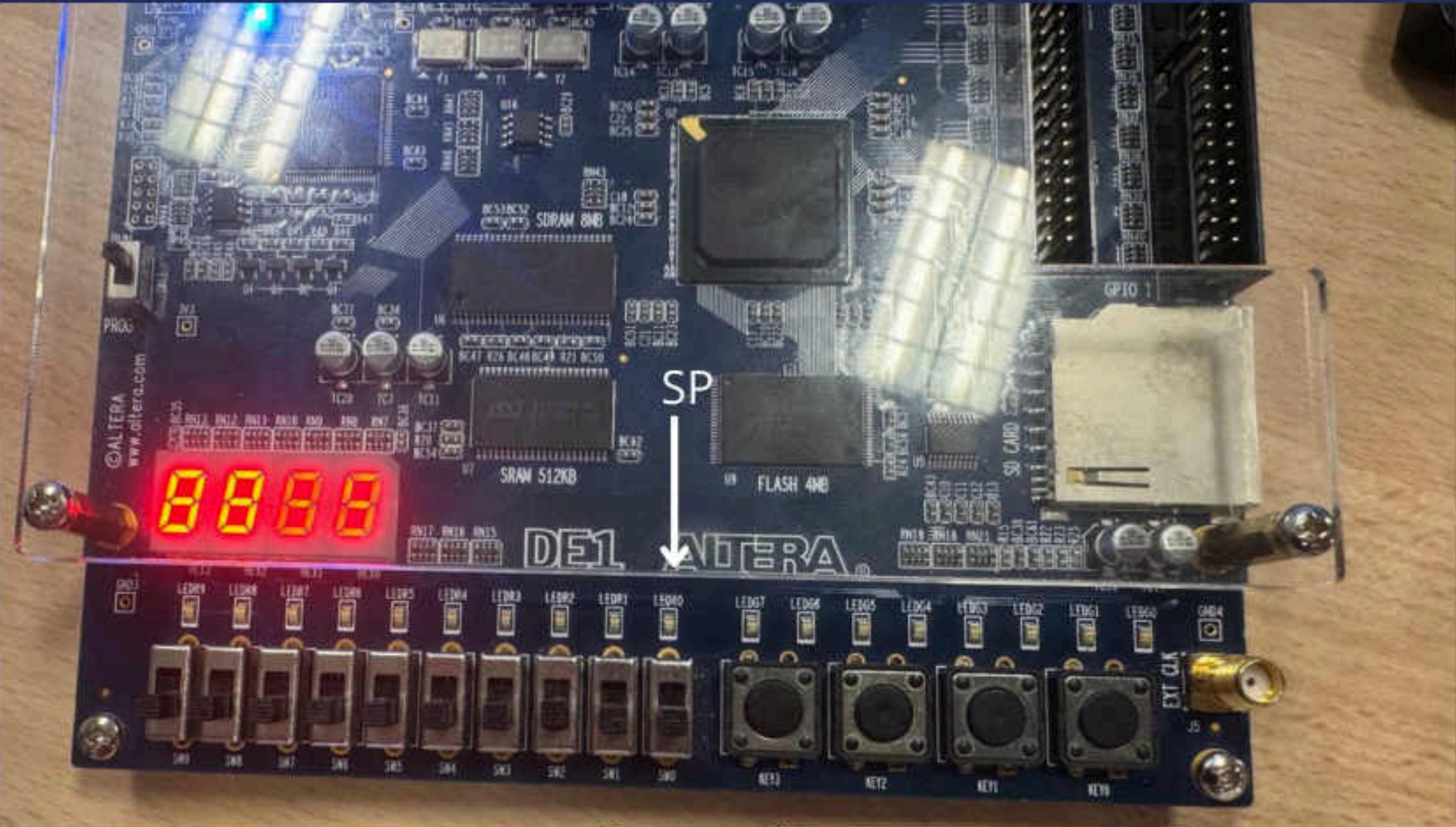
$SP = G.O.C.G.O.M.I_0.\bar{I}_1$

Table de karnaugh

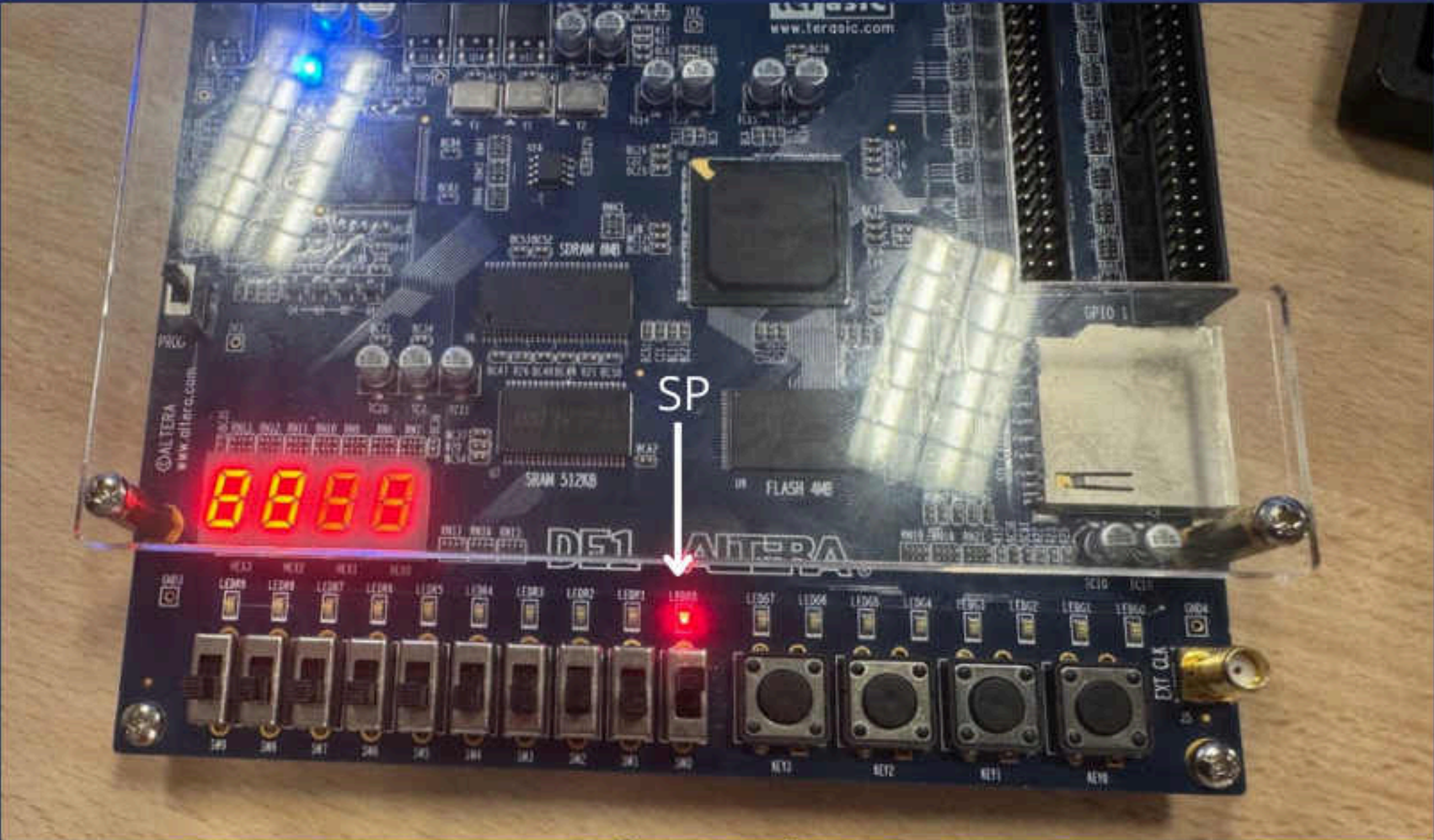
4. Bloc porte



4. Bloc porte

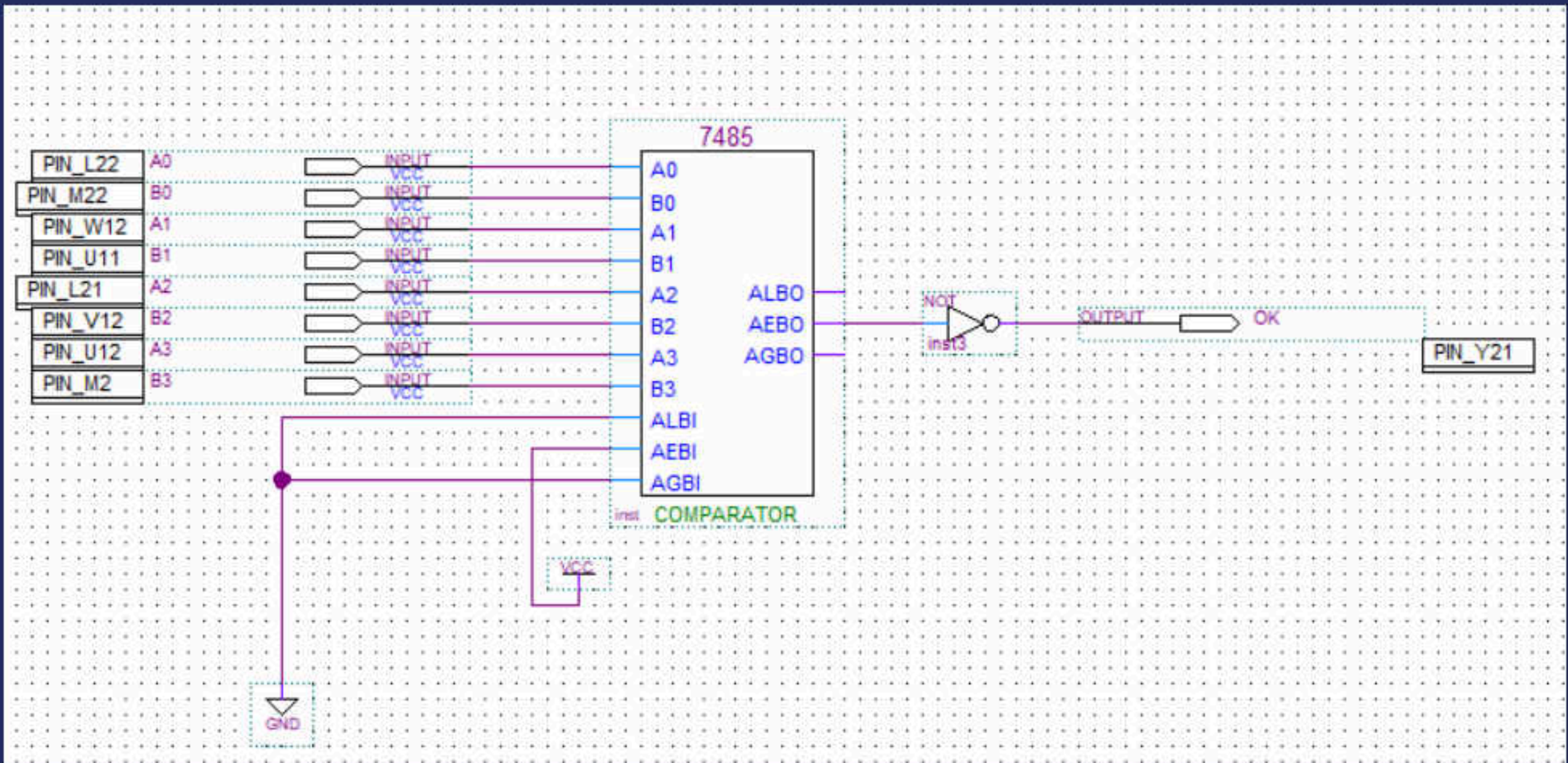
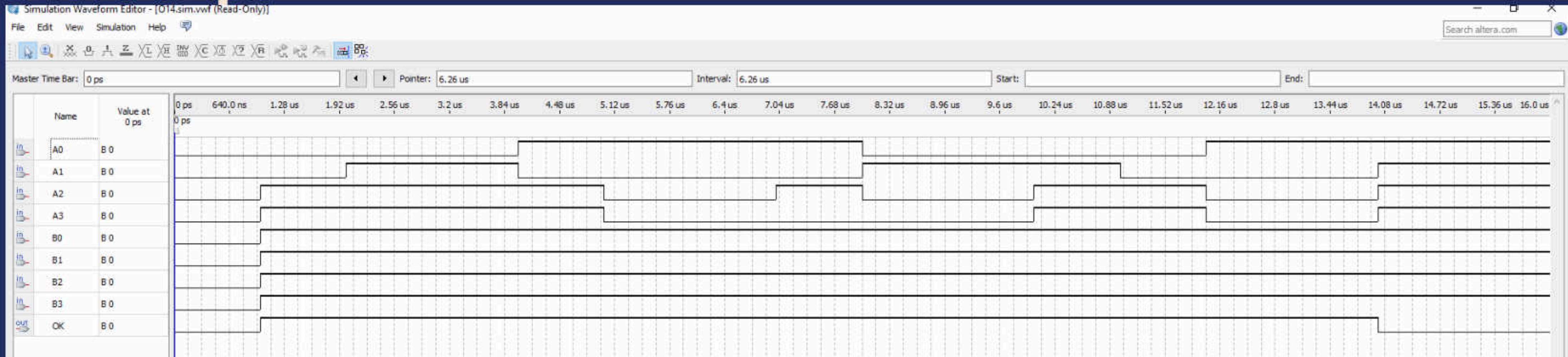


I1
I0
GOM
GOC

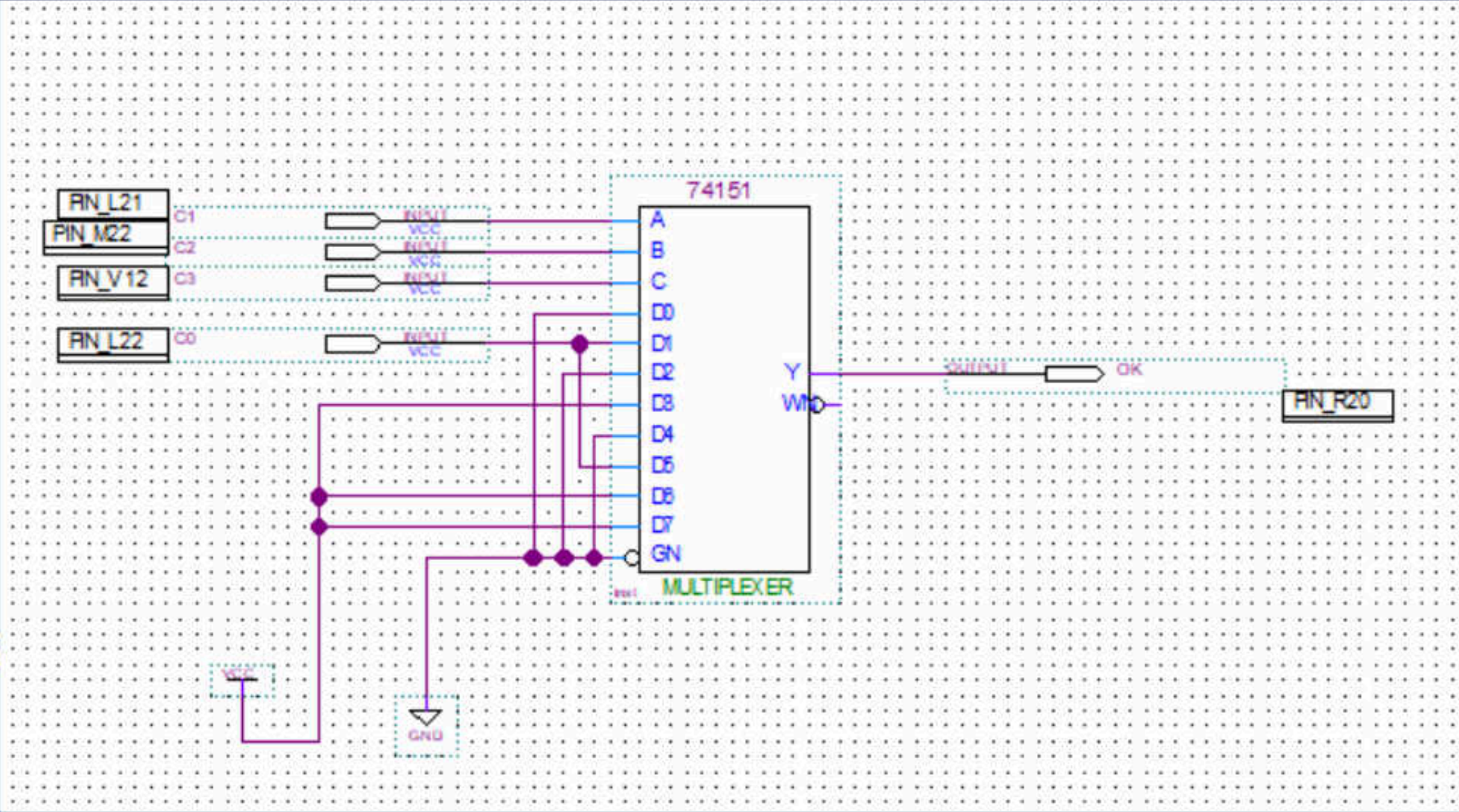
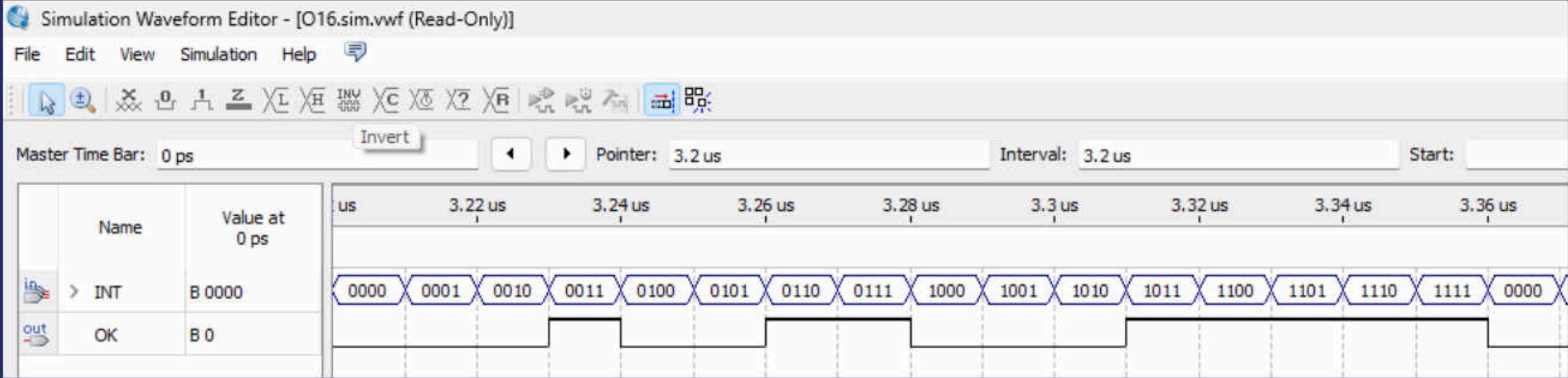


I1
I0
GOM
GOC

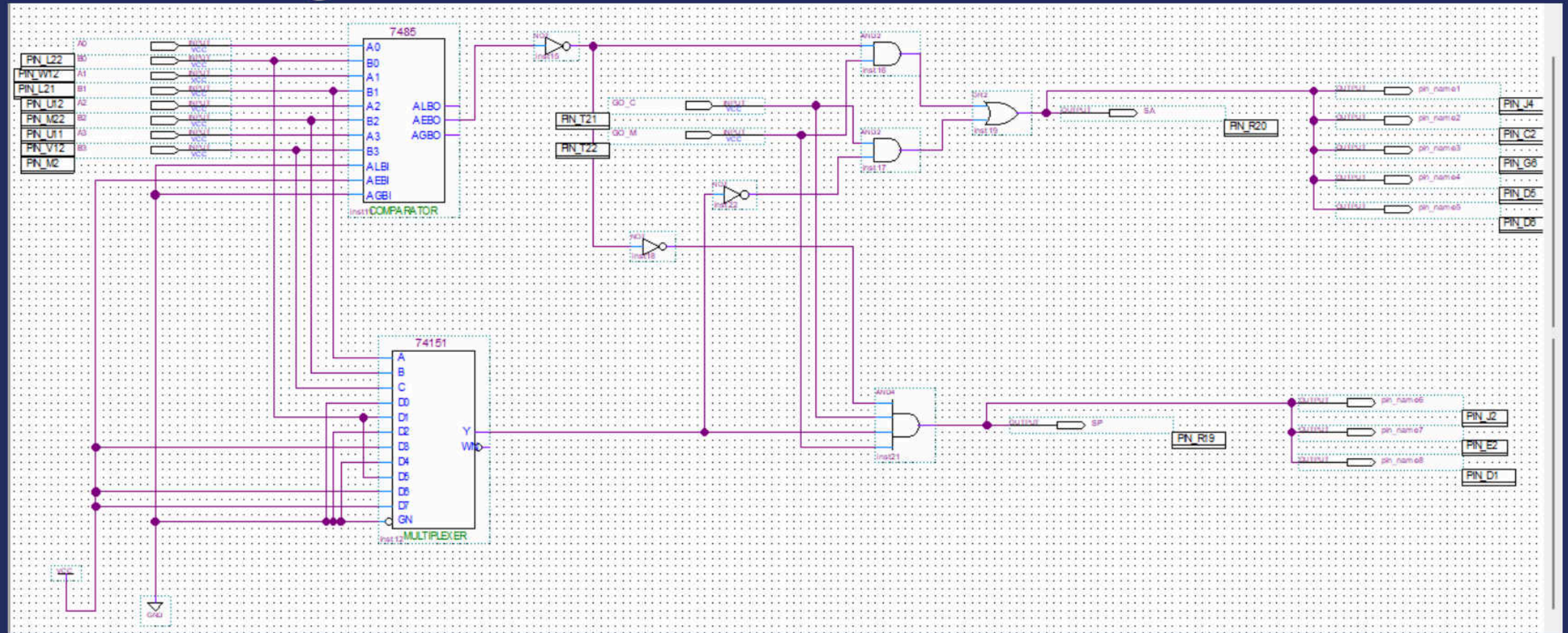
5. Bloc comparateur



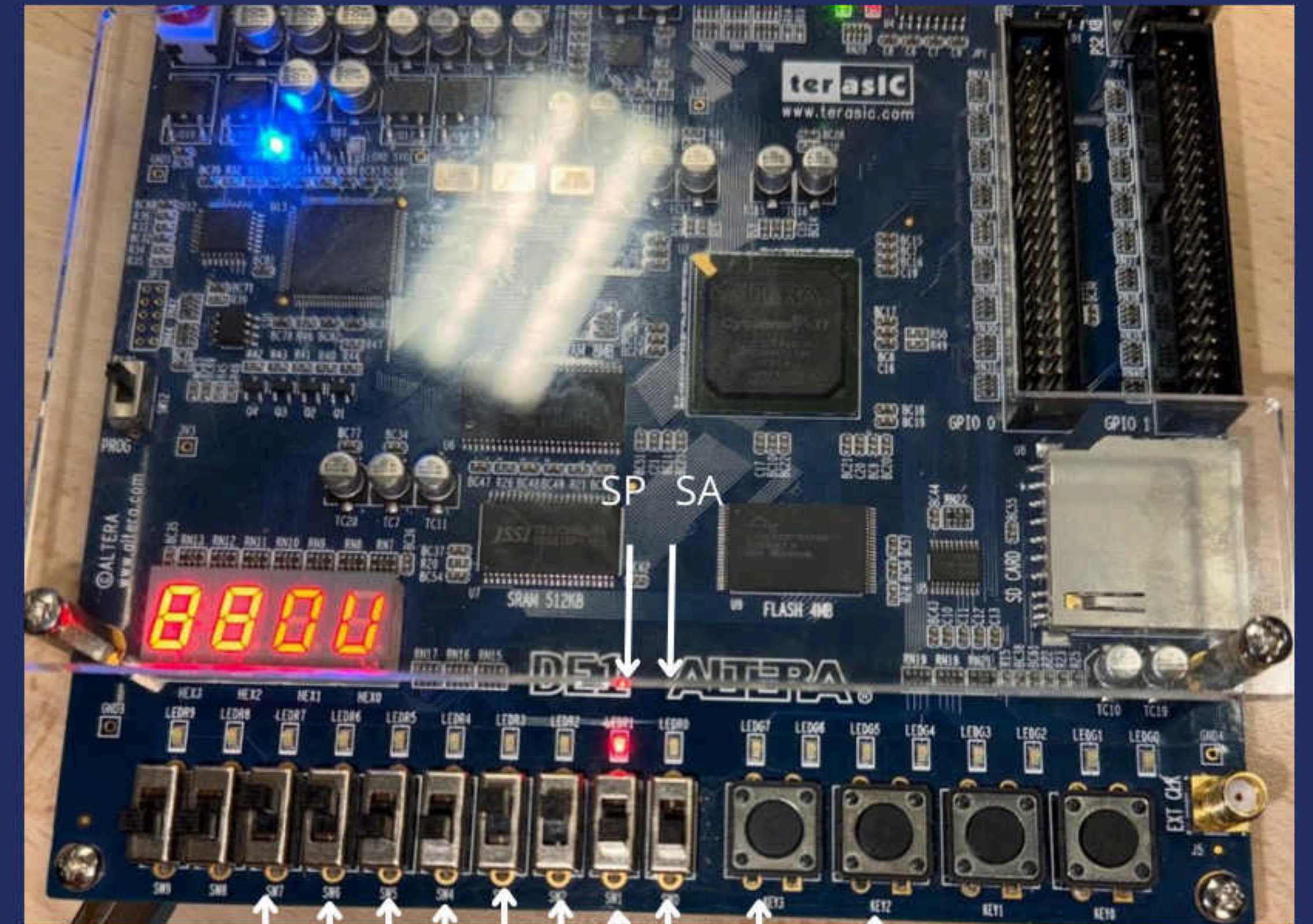
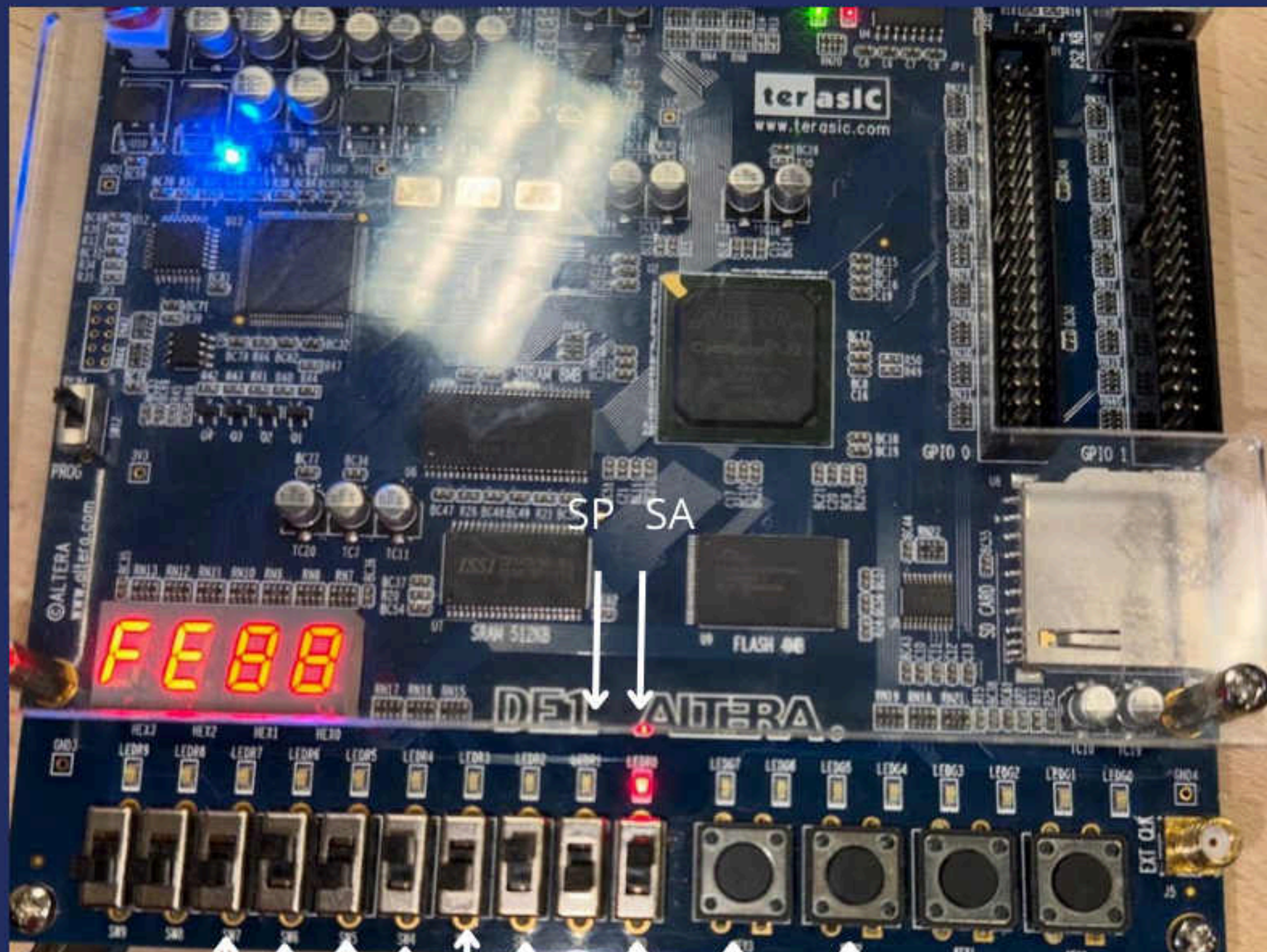
6. Bloc contrôle d'accès



7. Assemblage



7. Assemblage





8. Conclusion